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MILITARY SPECIFICATION

CLOTH, NONWOVEN, DISPOSABLE, CELLULOSIC

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of nonwoven scrim reinforced disposable cloth.

2. APPLICABLE DOCUMENT

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1136 - Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, or by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall be equal to or better than the sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.3), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2).

*3.3 Cloth, nonwoven. The cloth shall be a nonwoven laminate consisting of two plies of cellulosic tissue bonded to both sides of a synthetic reinforcement scrim. The color shall be any shade of white or yellow. The finished cloth shall conform to the requirements specified in Table I, when tested as specified in 4.5.

Table I - Physical requirements

Weight, oz/sq yd (maximum)	3.0
Yarns per inch, scrim (minimum)	
Machine direction	6.0
Cross direction	4.0
Breaking strength, lbs. (minimum):	
Machine direction	14.0
Cross direction	10.0
Tear resistance, grams (minimum):	
Machine direction	30.0
Cross direction	30.0
Seam efficiency, percent (minimum)	35.0
Water resistance, inches (minimum)	6.0
Air permeability, cu.ft/min (minimum)	25.0

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Table I - Physical requirements (cont'd)

Flame resistance	
After glow, seconds (maximum):	
Machine and cross directions	3.0
Char length, inches (maximum):	
Machine and cross directions	3.0

3.4 Length, width and put-up. The cloth shall be put up on rolls as specified in PPP-P-1136. The length and width shall be as specified (see 6.2).

3.5 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.6 Workmanship. The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements as specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected as specified in 4.3.2 for compliance with construction, workmanship, and end item testing.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

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4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. A certificate of compliance shall be submitted for the fiber requirements of 3.3.

4.3.2 Examination of the end item. The end item shall be examined in accordance with the classification of defects and at the inspection levels and acceptable quality levels (AQL's) set forth below.

4.3.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined, and defects clearly noticeable at normal inspection distance (3 feet) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the most serious defect shall be counted. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The acceptable quality level shall be 4.0 defects per 100 units for all defects listed. The sample size shall be in accordance with inspection level II of MIL-STD-105. The lot size shall be expressed in units of one yard each. The number of rolls selected shall be in accordance with Table II. An approximately equal number of yards shall be examined from each roll in the sample.

Defects

Any hole, cut or tear.
Any unbonded area, i.e., delamination of the non-woven outer layer from the scrim.
Any imbedded foreign matter.

4.3.2.2 Overall examination. Each defect shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size and the acceptance number shall be as shown in Table II.

Defects

Color not as specified
Overall uncleanness
Width less than specified
Objectionable odor
Not labeled or ticketed as specified

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Table II - Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up thru 1200 1/	3	0
1201 thru 3200	5	0
3201 thru 10,000	8	0
10,001 thru 35,000	13	0
35,001 thru 150,000	20	1
150,001 and over	32	2

1/ If the lot contains fewer than 3 rolls, each piece in the lot shall be examined.

4.3.3 Examination for length

4.3.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

* 4.4 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1136 to determine whether packaging, packing and marking comply with Section 5 requirements.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable and as listed in Table III shall be followed. The physical and chemical values specified in Section 3 shall apply to the results of the determination made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2 yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

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Table III - Test methods

Characteristic	Requirement paragraph	Test method
Weight	Table I	5041
Yarns per inch, scrim	Table I	5050
Breaking strength	Table I	5100
Tear resistance	Table I	5132
Seam efficiency	Table I	5110 1/
Water resistance	Table I	5514
Air Permeability	Table I	5450
Flame resistance	Table I	5903

1/ Except that the seam type shall be SSn or SSp with stitch type 504 or 505 as specified in FED-STD-751.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C, as specified (see 6.2 and 6.2.1).

* 5.1.1 Levels A and C. The cloth, put up as specified (see 3.4), shall be packaged in accordance with the applicable requirements of PPP-P-1136.

* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

* 6.1 Intended use. The cloth covered by this specification is intended for use in the fabrication of breathable low strength type disposable clothing that requires good protection against accidental flame.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Requirements for the first article (see 3.2).
c. Length and width required (see 3.4).
d. Levels of packaging and packing required (see 5.1 and 5.2).

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6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-1136 may be exercised in packaging.

* 6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Source of basic material Material conforming to the requirements of this specification may be obtained from Kimberly-Clark Corp., Neenah, Wisconsin 54956.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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